Exam 2 Recall

Gradients, directional derivatives, and tangent planes

1. Compute the gradient of a function of two or three variables.

2. Find a directional derivative from a vector or angle, and interpret its sign, magnitude, and units.

3. Find the directions and rates of steepest ascent and descent, and all directions in which the instantaneous rate of change is zero. Include the case where the gradient is zero.

4. Estimate gradients and sketch a gradient field from a contour plot. Compare directional derivatives using contour spacing and orientation.

5. Find a normal vector, normal line, and tangent plane to an explicitly or implicitly defined surface at a regular point. Find points where the tangent plane is parallel to a specified plane.

6. Find the linearization of a function of two variables at a point and use it to approximate nearby values.

7. For a hill given as a height function, find the angle of ascent or descent above horizontal in a specified direction.

Optimization

8. Find all critical points of a function of two variables, including points where a partial derivative does not exist but the function is defined. Classify them using the Second Derivative Test when applicable; use another method if the test is inconclusive.

9. Predict the locations and types of critical points from a contour plot. Check analytically when a formula is available.

10. State the Extreme Value Theorem and check its hypotheses. Find absolute extrema on a specified region by comparing all necessary interior, boundary, and corner candidates; parameterize straight or curved boundaries as needed. Justify that all necessary candidates have been included.

11. Translate a verbal optimization problem into an objective function, constraints, and the feasible domain.

12. Use Lagrange multipliers to find candidates for extrema of a function of two or three variables subject to one linear constraint.

13. Use Lagrange multipliers to find candidates for extrema of a function of two or three variables subject to one nonlinear constraint. Explain the gradient condition geometrically, and check constraint points where its gradient vanishes or does not exist.

14. Determine which constrained candidates give actual maxima or minima. Check whether extrema exist, including on unbounded constraint sets, and justify claims of nonexistence.

Double integrals

15. Interpret a double integral as signed volume and distinguish it from geometric volume. Evaluate an iterated integral over a rectangle.

16. Sketch a region from integration bounds. Write bounds in both orders for a region described by curves, inequalities, or a diagram, splitting the region when necessary.

17. Set up and evaluate double integrals over nonrectangular regions. Reverse the order of integration when it makes evaluation easier.

18. Use double integrals to find area and volume between surfaces. Sketch the solid represented by a given volume integral.

19. Describe a region in polar coordinates and convert a Cartesian double integral to polar coordinates, including the area element. Choose Cartesian or polar coordinates based on the region and integrand.

20. Use polar coordinates to evaluate a double integral or a volume involving spheres, cylinders, or paraboloids. Determine radial and angular bounds from the geometry.

21. Find the average value of a function over a planar region, including an average distance. Identify the region whose area belongs in the denominator.

22. Determine when symmetry or separation into a product of one-variable integrals simplifies a double integral. Justify the required conditions or give a counterexample.

Vector fields and line integrals

23. Sketch a vector field or match a formula to its plot.

24. Predict the sign of a vector line integral from a field diagram and an oriented curve.

25. Evaluate a scalar line integral and use it to find the mass of a wire with variable linear density.

26. Evaluate a vector line integral in two or three dimensions.

27. Explain how reversing orientation or making a regular one-to-one change of parameter affects scalar and vector line integrals.

Conservative fields and potential functions

28. Test whether a vector field is conservative on its stated domain.

29. Find a potential function in two or three dimensions.

30. State the Fundamental Theorem for Line Integrals and its hypotheses. Evaluate a line integral using a potential, and compare with direct integration of the same integral.

31. Explain the relationships among conservativeness, path independence, and integrals around closed curves.

32. Use path independence to compare integrals along paths with the same endpoints. Evaluate the line integral of a conservative field along a complicated path by using a potential function or replacing the path with a simpler one having the same endpoints in the field's domain.

Green's theorem

33. State Green's theorem and its hypotheses.

34. Use Green's theorem in either direction to convert between a closed line integral and a double integral, accounting for boundary orientation.

35. Evaluate an open-curve integral by adding a closing segment, applying Green's theorem, and subtracting the added contribution.

36. Find the area enclosed by a parameterized curve using Green's theorem. Check the orientation and that the parameter interval traces the intended boundary exactly once.

Curl and divergence

37. Compute curl and divergence. Explain their local geometric interpretations.

38. Infer the sign of divergence and the direction of curl from a suitable vector-field diagram. Explain the local behavior supporting each conclusion.

39. Use the divergence of a curl to rule out a proposed field being a curl, stating the differentiability assumptions.

40. Determine whether zero curl, zero divergence, or both force a field to be constant. Justify each conclusion or give a counterexample.